

A pathophysiological inflection point in Duchenne muscular dystrophy uncovered by myofibre density analysis: implications for therapeutic strategies

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Myofibre Density (MFD), a simple count of myofibres per mm², reveals a phase of rapid myofibre loss in steroid-naïve Duchenne muscular dystrophy (DMD), with a histopathological critical period around age 6.

Background and Rationale

Early DMD symptoms emerge at 3–5 yrs, yet prior histological work suggested little change until ~age 6. Preliminary screening showed strong inverse age correlation for **myofibre density (MFD)**, a density metric distinct from classical area measures.

QUESTION

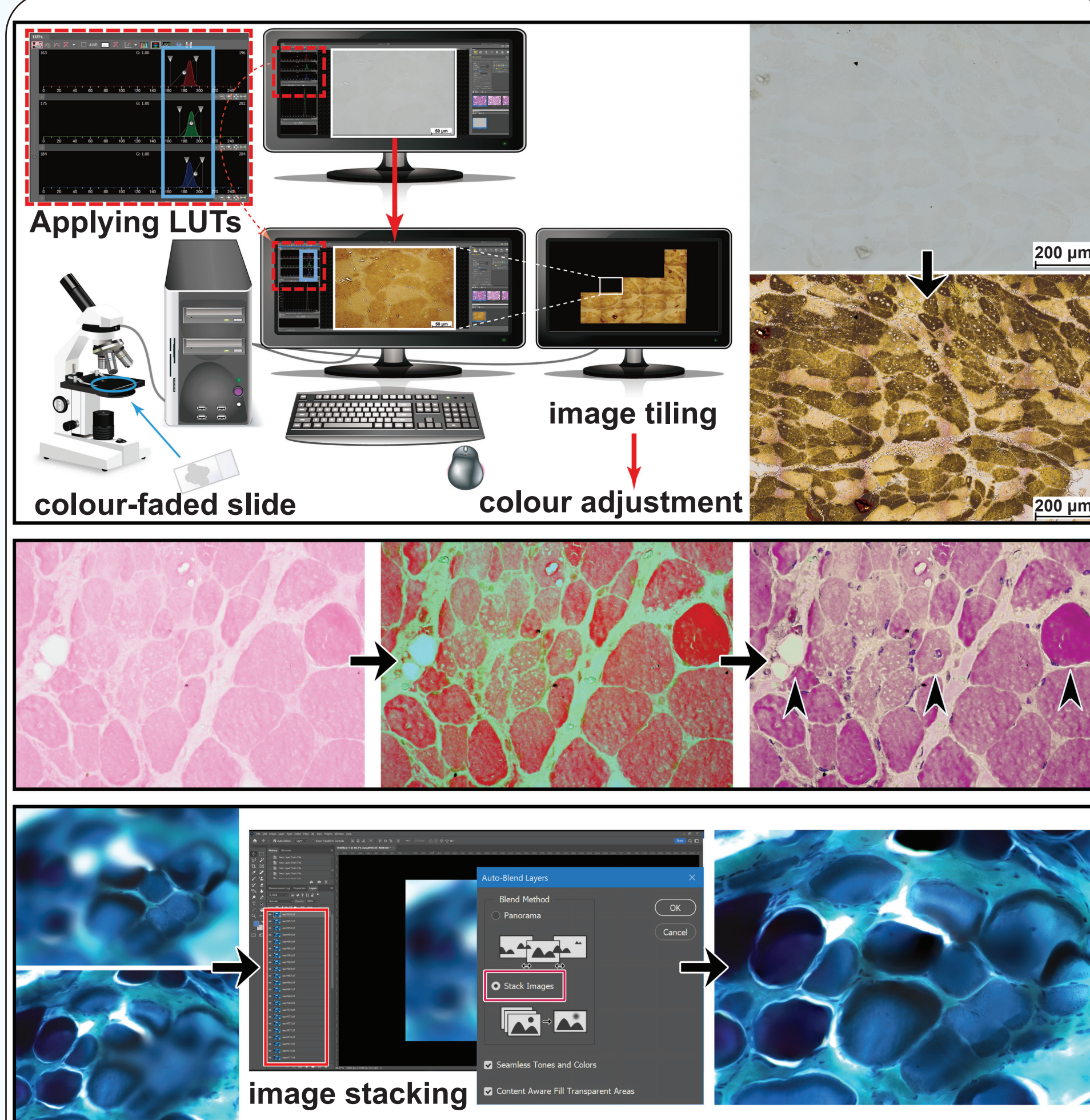
Can MFD bridge this clinicopathological gap by revealing an unrecognised age–MFD trajectory in steroid-naïve DMD?

Cohort and Sample Flow

46 total → 38 quality-check → 35 age screening

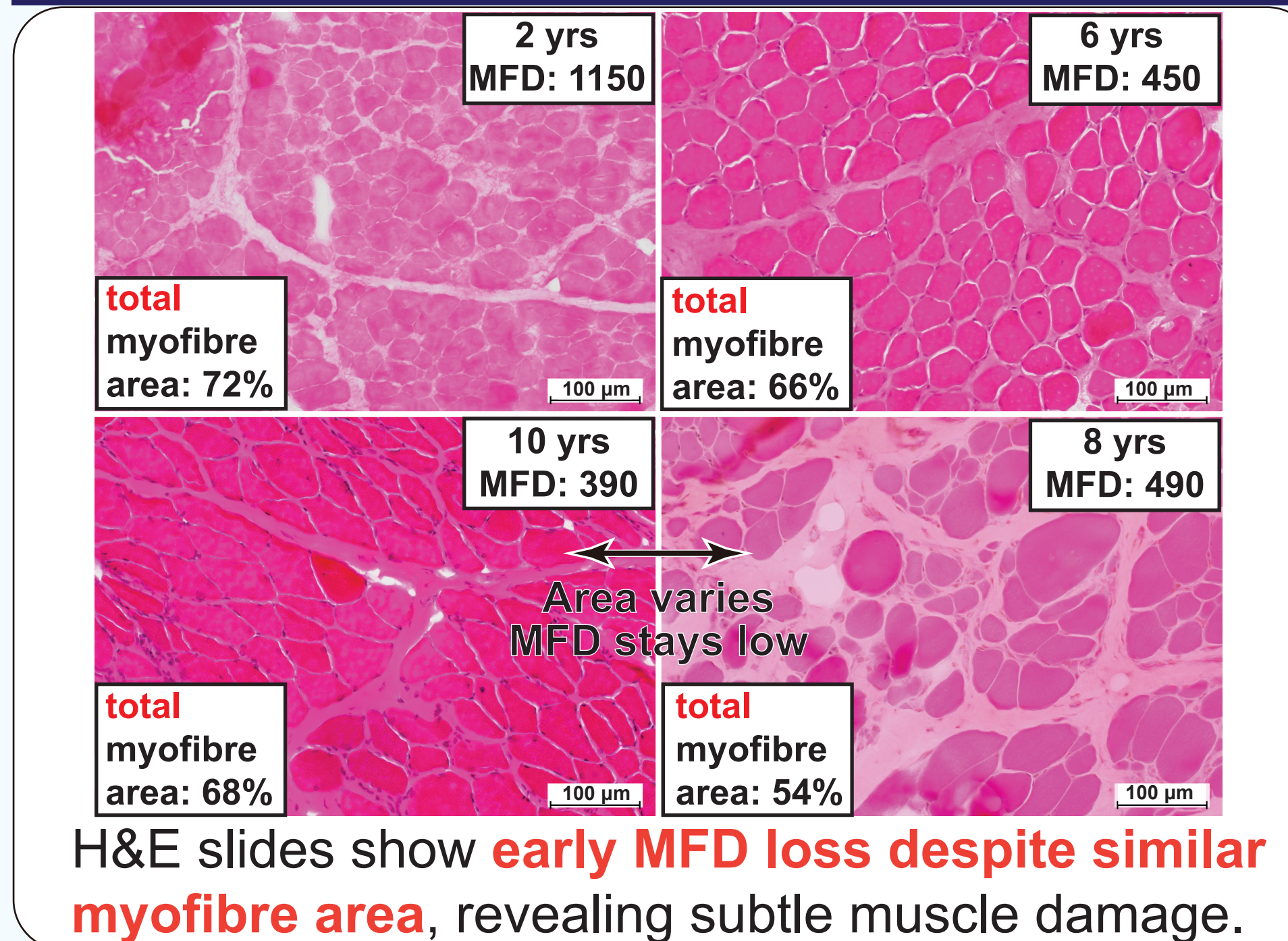
Steroid-naïve DMD quad-biopsy slides collected in 1973–1984. Main segmented regression included age 1–11 years; 11–16-year samples (n = 3) were excluded after the age-distribution screening.

Digital Restoration and Quantification



Faded archival sections were restored with Look-Up Tables (LUTs) adjustment and Focus Stacking, then tiled into high-resolution composite images for subsequent semi-quantitative analysis.

Representative Histology



H&E slides show **early MFD loss despite similar myofibre area**, revealing subtle muscle damage.

Analytical Framework

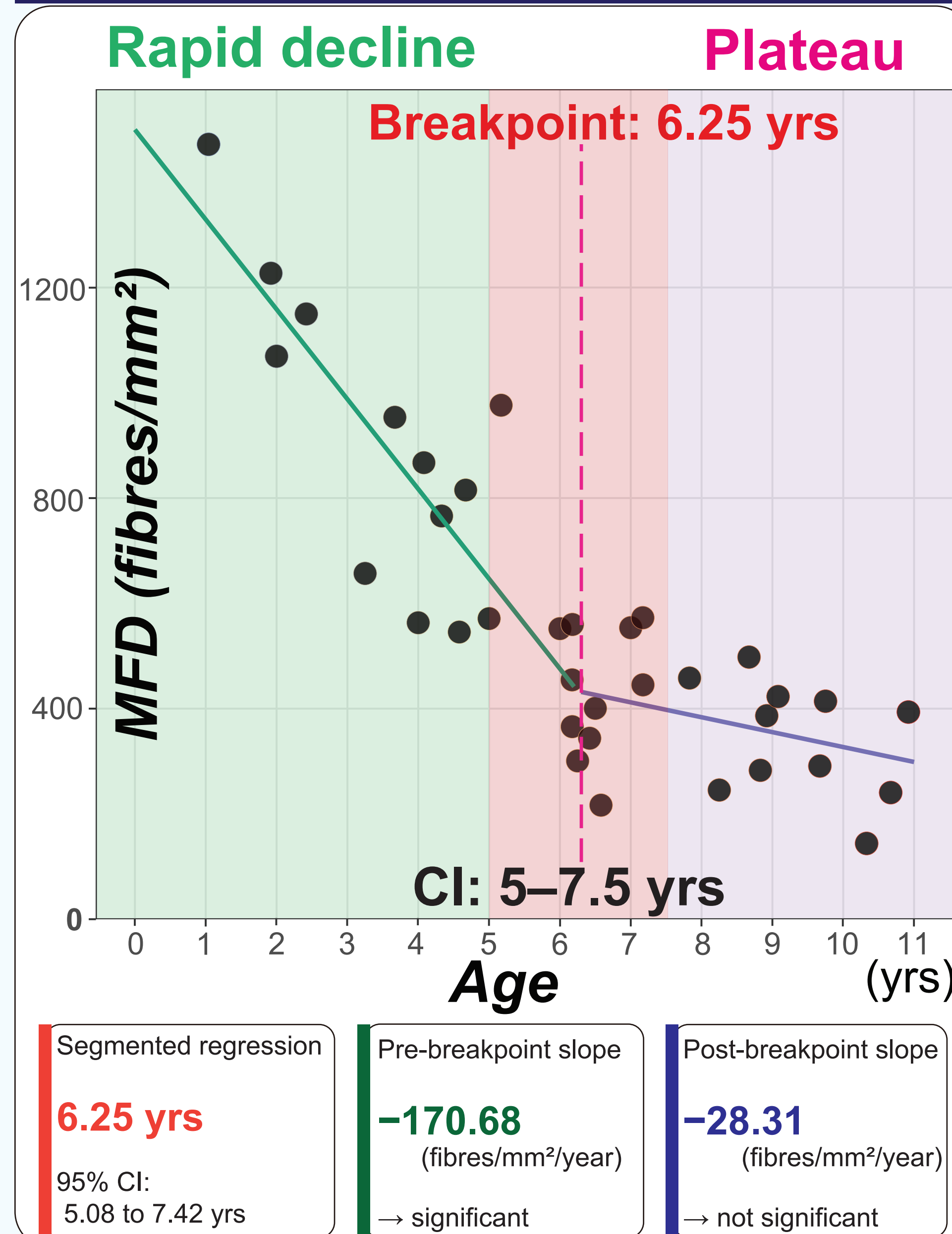
Primary measure

Age-dependent MFD breakpoint estimated with segmented regression and Bayesian modelling.

Robustness and secondary analyses

Monte Carlo **breakpoint-recapture simulation**; logistic MFD **age-predictive cutoffs**; variance analysis across **classical histological variables**.

Age–MFD Dynamics



Bayesian Validation and Power Analysis

Bayesian breakpoint

6.37 yrs

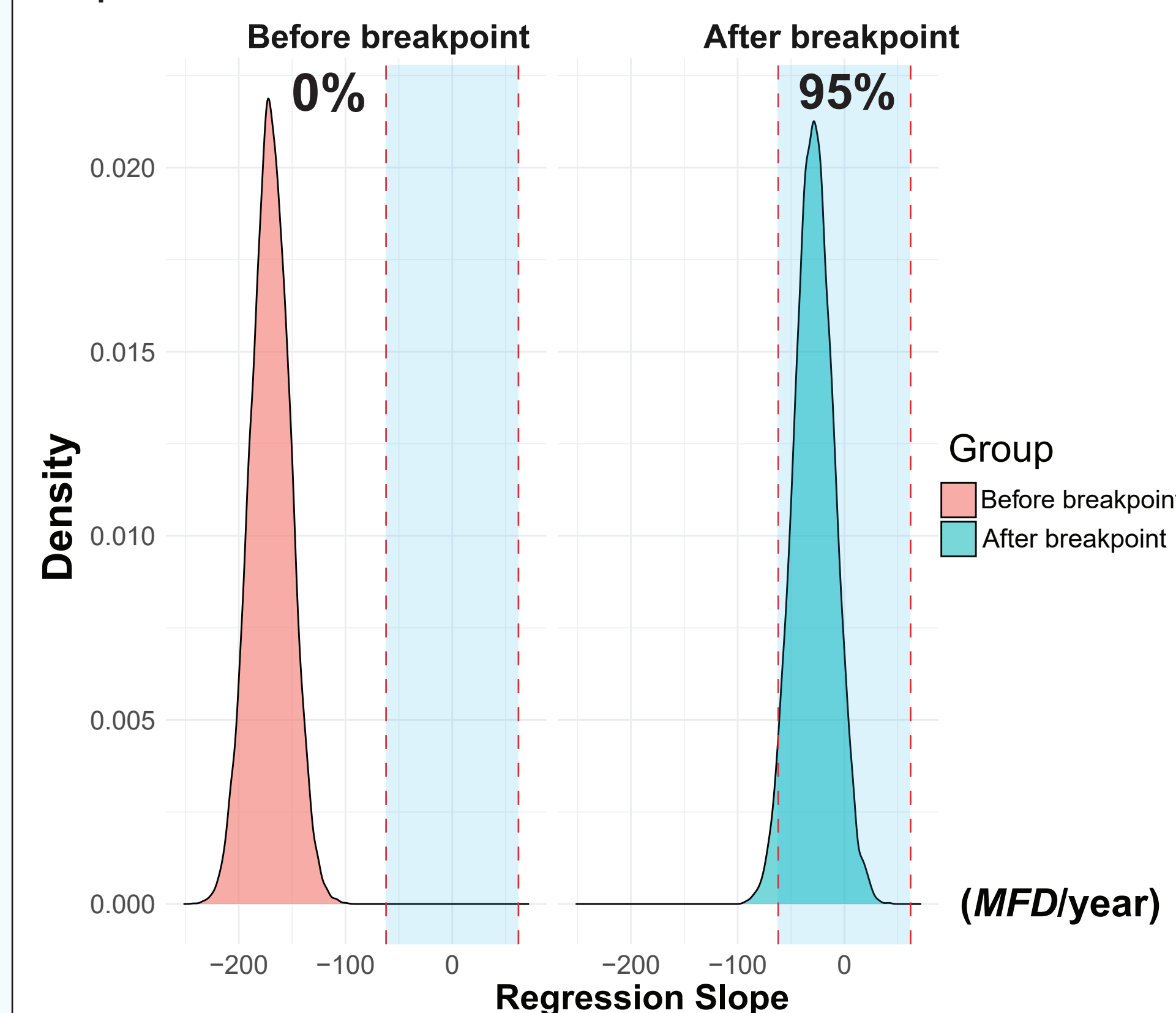
95% CrI (credible interval): 5.24 to 7.66 yrs

10,000-run Monte Carlo

79.47%

Recaptured breakpoint within +/-1.25 yrs (n = 35)

Bayesian analyses confirmed steep pre-breakpoint decline and a post-breakpoint slope practically equivalent to zero within the **ROPE***.



*Region of Practical Equivalence; slope range treated as practically zero.

Age-Predictive MFD Thresholds

Logistic modelling translated MFD values into probabilities of being younger than the 6.25 yrs.

Early zone >595.51 fibres/mm²

>80% probability of age being before 6.25 yrs

Transition zone 426.36 to 595.51 fibres/mm²

20–80% probability; individual judgement

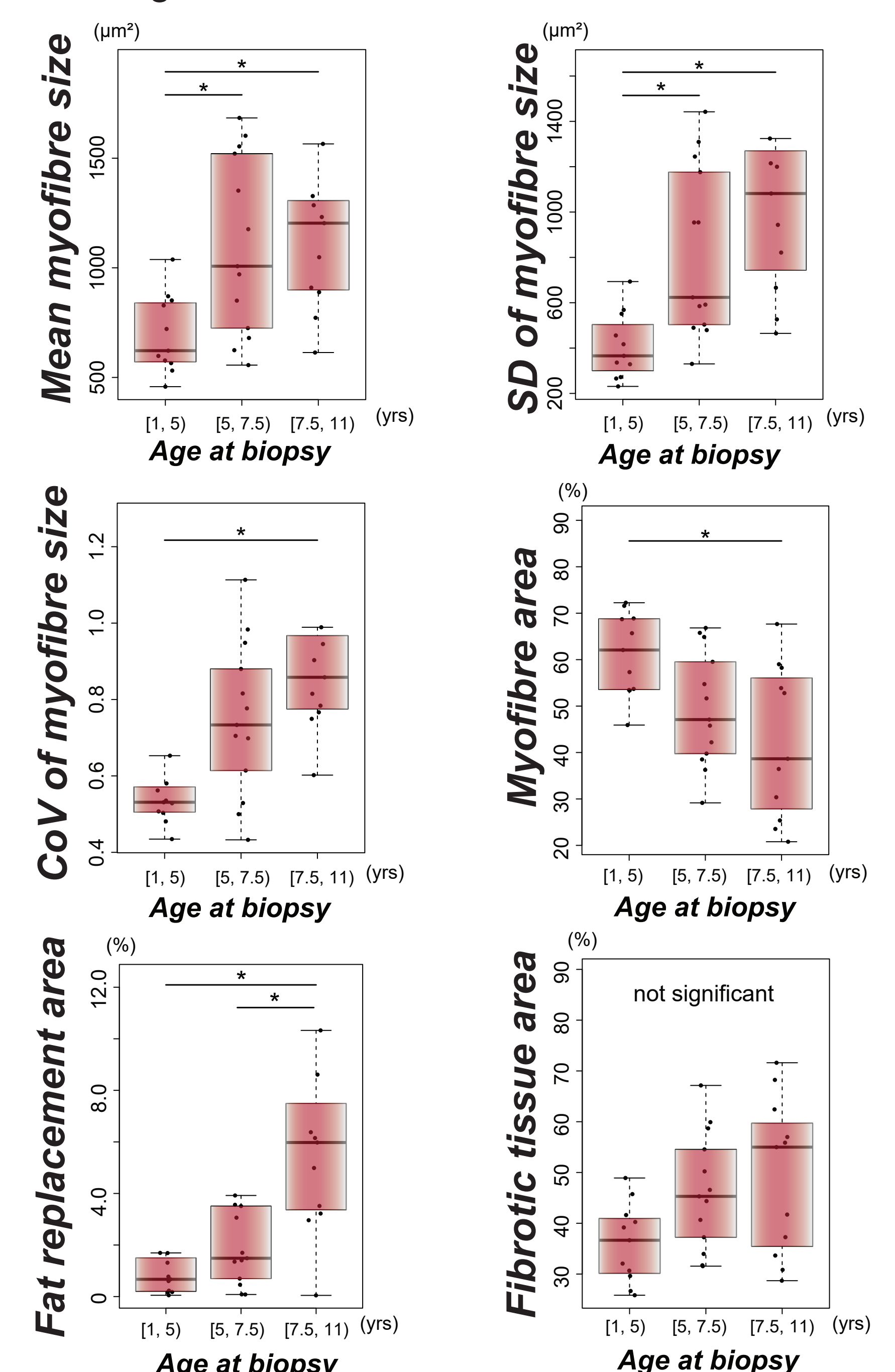
Late zone <426.36 fibres/mm²

<20% probability of age being before 6.25 yrs

These cut-offs are hypothetical and need prospective clinical validation in **biceps-biopsy** cohorts.

Cross-Parameter Validation

Breakpoint-derived age bands show coherent shifts in classical histological variables, supporting the biological relevance of the MFD threshold.



Conclusion

- “**MFD**” reveals a previously unrecognised phase of **rapid myofibre loss** lasting up to **around age 6** in steroid-naïve DMD.
- This simple metric may add pathological evidence to inform the ongoing debate on the **optimal timing of steroid initiation**.

Key references

1. Yamakado T et al. *Neuropathol Appl Neurobiol*. 2026;52(2):e70068. 2. Birnkrant DJ et al. *Lancet Neurol*. 2018;17(3):251–267. 3. Pevereli L et al. *Neurology*. 2015;85(21):1886–1893. 4. Desguerre I et al. *J Neuropathol Exp Neurol*. 2009;68(7):762–773.

Ethics/COI/funding

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QR code:
Full paper, WMS poster,
supplementary data and
reproducible analysis code.

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